Discussion and Conclusion: What the Artifact Establishes—and What It Does Not

Principal findings: a claim-calibrated curriculum artifact I

Active Skillference establishes a bounded engineering result: a contested formal subject can be represented as an inspectable prerequisite graph whose structural, quantitative, scholarly, and delivery contracts are checked separately. The curriculum graph is a validated DAG; learner-facing computed values are resolved from tested kernels; citations are assigned source roles; and generated figures, manuscript references, and SkillTree payloads are audited as connected artifacts. These are properties of this checkout and its build process, not evidence that the selected order is optimal for every learner or that the underlying scientific framework is empirically confirmed.

Interpretation: infrastructure rather than intelligence or personalization I

The mathematical examples and the curriculum infrastructure operate at different levels. The kernels implement small, model-dependent demonstrations of inference and policy evaluation. The curriculum layer orders and assesses concepts. The provenance layer records how values and sources enter the learner-facing artifact. None of these layers is an intelligent tutor, a general-purpose reasoning system, or an adaptive learner model. In particular, the term *agent* in the mathematical examples refers to a formal model with specified states, observations, preferences, and policies; it should not be read as a claim about consciousness, general intelligence, autonomy, or human-like learning.

Interpretation: infrastructure rather than intelligence or personalization I

The related-work literature supports this separation. Active Inference formulations can connect belief updating, action selection, learning, and epistemic behavior, but their predictions depend on the generative model, preferences, approximation, and policy set (Rao and Ballard 1999; Friston 2013, 2019; Ramstead et al. 2018). The supplement therefore keeps the VFE/EFE derivations, sign conventions, and preference assumptions visible instead of treating one computed policy comparison as a universal account of intelligent behavior. Learner-state models such as knowledge tracing and Q-matrix cognitive diagnosis belong to a separate adaptive-learning evidence layer and are cited here as comparison boundaries, not as components of this artifact (Piech et al. 2015; Tao et al. 2024).

Implications: future adaptive learning systems and open-science interoperability I

The delivery contract is useful precisely because it does not hide ownership boundaries. The exporter creates a SkillTree-compatible Project -> Subject -> Skill profile with learning-path dependencies and quiz associations. The local dashboard exposes generated artifacts and opens official subject and quiz routes, while SkillTree remains the owner of progress, ranking, scoring, completion, and authentication. This makes the repository a candidate foundation for a later adaptive study, not an adaptive system in the present release.

Implications: future adaptive learning systems and open-science interoperability I

A genuinely adaptive extension would require learner response data, a validated model of learner state, a transparent rule for selecting the next skill, item-level measurement, privacy and consent controls, and prospective evaluation against a preregistered comparator (Doignon and Falmagne 1985; Fu and Fang 2025; Design-Based Research Collective 2003; Liu and Khalil 2023; Piech et al. 2015; Tao et al. 2024). Knowledge tracing and Q-matrix cognitive diagnosis are comparison literatures for those stronger commitments, not components of the present export. Provenance and graph validation would remain necessary infrastructure, but they would not by themselves establish personalization quality or learning effectiveness.

Implications: future adaptive learning systems and open-science interoperability I

The community contribution should be read at the same bounded level. The Active Inference Institute's public website repository presents a public resource hub for learning, research, participation, and contribution, with public repositories and a documented boundary between internal inputs and verified public projections (Active Inference Institute 2026). Active Skillference contributes a complementary artifact to that open-science environment: it packages a prerequisite graph, kernel-backed examples, source-role audit, and SkillTree handoff into a rerunnable educational release. The relationship is interoperable and community-facing, not institutional: this repository does not claim to be an Institute product, an official curriculum, or an endorsement of any contributor or public pathway.

Limitations and planned empirical evaluation I

The release state is machine-readable. As of 2026-07-31, `tasks.yaml` records 78 roadmap tasks: 74 completed milestones and 4 open items across metadata, skilltree, validation. `TODO.md` is rendered from the same source. fig. 1 shows the current distribution of open versus completed work by surface and release class, keeping release-readiness language tied to the validator state that drives the roadmap.

Limitations and planned empirical evaluation I

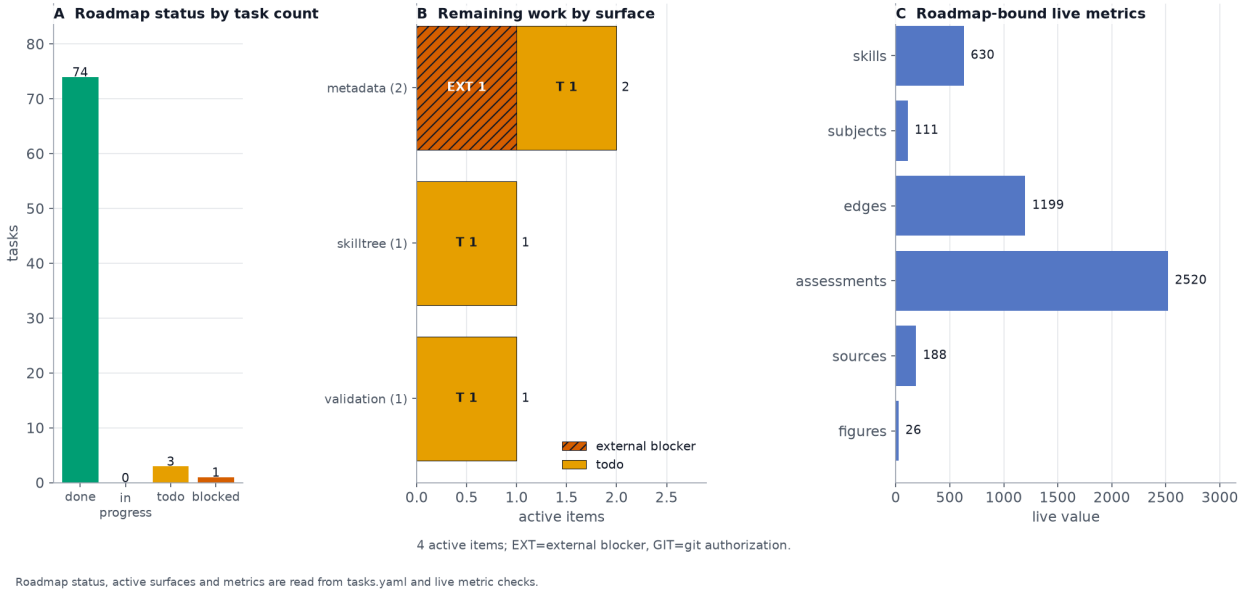


Figure 1: Roadmap status by artifact surface generated from `tasks.yaml` and the live metrics stored in the roadmap source data. The first panel counts roadmap items by status, the middle panel groups remaining open or blocked work by surface and release class with direct EXT/GIT segment labels, hatching, and surface totals, and the final panel reports current curriculum, assessment, scholarship, and figure metrics. The figure separates external blockers from git-authorization hygiene using machine-readable roadmap data, so release-readiness text must agree with the same source.

Limitations and planned empirical evaluation I

The limitations are therefore substantive rather than cosmetic. The kernels cover fixed, small discrete teaching cases and bounded predictive-coding or planning demonstrations; their conclusions depend on the stated model, preferences, inputs, and approximation regime. The runtime depends on SkillTree for live progress and quiz state, and this release contains no learner-response data, learner-state inference, retention measure, transfer measure, efficacy estimate, or differential-benefit analysis. The planned pilot described in sec. 13 and the fuller boundary analysis in sec. 19 define the evidence needed before those claims can be made.

Conclusion: a bounded, reproducible contribution I

The strongest conclusion is narrower and more useful: Active Skillference makes the construction, provenance, and delivery of a formal Active Inference curriculum more inspectable, reproducible, and auditable. It establishes an infrastructure artifact and an open-science-compatible handoff surface; it does not establish intelligent tutoring, personalization, learner improvement, or empirical confirmation of the Free Energy Principle.